⚛ ATOMIC STRUCTURE — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Modern Chemistry & Dalton Atomic Theory

- Antoine Lavoisier ko 'Father of Modern Chemistry' maana jata hai. Combustion mein oxygen ki role ko establish karne, oxygen (1778) aur hydrogen (1783) ko name karne aur chemical nomenclature ko reform karne mein unka major contribution tha. [Q1]
- Dalton atomic theory ke according matter atoms se bana hai. Is theory ne law of conservation of mass, law of constant composition aur law of multiple proportions ko explain kiya; radioactivity ko explain nahi kiya. [Q2]

2. Early Atomic Models — Thomson & Rutherford

- J.J. Thomson ne atom ka pehla structural model propose kiya — positively charged sphere ke andar electrons embedded hone wala 'plum pudding' model. Baad mein Rutherford aur Bohr ke models ne isse replace kiya. [Q3]
- Atomic nucleus ki discovery Ernest Rutherford se associate ki jati hai. Gold-foil / Geiger–Marsden experiment ke observations se clear hua ki atom ka almost sara positive charge aur mass ek bahut chhote dense centre mein concentrated hota hai. [Q11]

3. Atom, Nucleus & Basic Constituents

- Atom ka centre wala positively charged dense part nucleus kehlata hai. Nucleus mainly protons aur neutrons se milkar bana hota hai; in dono ko collectively nucleons kehte hain. [Q4, Q5, Q6]
- Electron nucleus ke andar nahi hota; atom ke surrounding region/orbitals mein milta hai. Neutral atom mein number of protons = number of electrons hota hai, isliye net charge zero hota hai. [Q7]
- Basic atomic picture mein protons + neutrons nucleus mein concentrated hote hain, jabki electrons nucleus ke around shells/orbitals mein occupy karte hain. [Q8]
- Proton, neutron aur electron atom ke main constituent particles hain. Photon electromagnetic radiation ka quantum hai, atom ka structural constituent nahi. [Q9]
- Element ek aisa pure substance hai jisme same atomic number/type ke atoms hote hain; compounds aur mixtures mein ek se zyada kinds of atoms/elements involve ho sakte hain. [Q10]

4. Discovery Timeline — Electron, Proton, Neutron & π-Meson

- Exam chronology: electron — J.J. Thomson (1897) → proton — Rutherford se associated (source: 1919) → neutron — James Chadwick (1932) → π-meson — Cecil Powell (1947). Isliye order III → IV → II → I hai. [Q12]

- James Chadwick ne 1932 mein neutron discover kiya. Neutron electrically neutral particle hai aur nucleus ka important constituent hai. [Q13]

⚠ EXAM TRAP — Proton Discovery — Exam Chronology vs Historical Detail — Q12

▸ Source chronology Rutherford/proton ko 1919 ke saath use karti hai. Exam sequence ke liye electron 1897 → proton → neutron 1932 → π-meson 1947 yaad rakhein. [Q12]

▸ Goldstein ki canal rays (1886) positive rays ka earlier observation tha; Rutherford ko proton identification/nuclear hydrogen context se associate kiya jata hai. [Q12]

5. Subatomic, Fundamental & Antiparticle Concepts

- Deuteron, deuterium nucleus hai jisme 1 proton + 1 neutron hota hai. Ye composite nucleus hai; electron/proton/neutron jaise basic constituent-particle list se alag category mein aata hai. [Q14]
- Positron electron ka antiparticle hai: mass electron ke equal hota hai, lekin charge +1e hota hai. Electron ka charge −1e hota hai. [Q15]
- Source matching: neutrino → historically 'zero mass'; quark → fractional charge (+2/3e ya −1/3e); positron → spin 1/2; phonon → integral-spin bosonic quasiparticle. [Q16]
- Quarks fundamental particles hain; proton aur neutron composite hadrons hain jo quarks se bane hote hain. [Q17]

⚠ EXAM TRAP — Deuteron & Neutrino — Source Simplification — Q14, Q16

▸ Deuteron technically ek composite subatomic nucleus/particle hai; source ka intent ise elementary/basic constituent particle se distinguish karna hai. [Q14]

▸ Neutrino ko old texts mein zero-mass likha milta hai, lekin modern physics neutrino oscillations se tiny non-zero mass establish karti hai. Exam mein source-era wording ko context ke saath dekhein. [Q16]

6. Higgs Boson — "God Particle" & Mass

- Higgs boson ko popular media mein 'God Particle' kaha gaya. Ye Higgs field se related elementary boson hai, aur Higgs mechanism elementary particles ko mass acquire karne ko explain karta hai. [Q18, Q19, Q20]
- Higgs boson discovery ka real significance ye samajhna hai ki elementary particles mass kyun rakhte hain. Matter teleportation ya better nuclear-fission fuel iska direct implication nahi hai. [Q21]

⚠ EXAM TRAP — Higgs Boson — Discovery Year — Q19–Q21

▸ Higgs-like boson discovery CERN ke ATLAS/CMS experiments ne July 2012 mein announce ki; 2013 Nobel Prize Higgs mechanism theory ke liye Higgs aur Englert ko mila. Source ki '2013 proved' wording ko exact discovery date na samjhein. [Q19, Q21]

▸ 'God Particle' popular nickname hai; scientific name Higgs boson hi hai. [Q18, Q19, Q20]

7. Charge Map — Neutron, Neutrino, Alpha & Beta

- Neutron electrically neutral hota hai. Given charged-particle comparisons mein proton +1e, electron −1e aur alpha particle +2e charge carry karta hai. [Q22]
- Neutrino ka electric charge zero hota hai aur mass extremely small but non-zero maana jata hai. Positron +1e, electron −1e aur alpha particle +2e hota hai. [Q23]
- Alpha particle basically helium-4 nucleus hota hai: 2 protons + 2 neutrons. Isliye iska mass helium nucleus ke nearly equal hota hai aur charge +2e hota hai. [Q24, Q25, Q28]
- Ordinary hydrogen ka most common isotope protium (^1H) nucleus sirf 1 proton rakhta hai aur neutron zero hota hai. [Q26]

- Standard β^- particle high-speed electron hota hai aur negative charge carry karta hai; X-rays aur gamma rays neutral electromagnetic radiation hain. [Q27]

⚠ EXAM TRAP — Beta Particle — Beta-minus vs Beta-plus — Q27

▸ Exam mein 'beta particle carries negative charge' usually β^- electron ko refer karta hai. β^+ decay mein positron emit hota hai, jiska charge positive hota hai. [Q27]

8. Orbitals & Quantum Numbers

- Aufbau principle orbital-filling order govern karta hai: electrons pehle lower-energy orbitals fill karte hain, phir higher-energy orbitals. [Q29]
- Magnetic quantum number m_l orbital ki spatial orientation batata hai. Principal quantum number n shell/energy level, azimuthal quantum number l subshell/orbital shape, aur spin quantum number m_s electron spin ko represent karta hai. [Q30]

9. Mass Number & Neutron Calculations

- Neutral atom mein electrons = protons. Agar electrons 18 aur neutrons 20 hain, to protons 18 honge; mass number A = p + n = 18 + 20 = 38. [Q31]
- Mass number nucleus ke protons + neutrons ka sum hai; electrons mass-number calculation mein count nahi hote. 2p + 2n wala atom ka A = 4. [Q32, Q33]
- $^{242}Pu_{94}$ ke liye neutron number $N = A - Z = 242 - 94 = 148$. [Q34]

10. Isotopes & Isoneutronic / Isotone Groups

- Isotopes same element ke atoms hote hain: atomic number / proton number same hota hai, neutron number different; isliye mass number different hota hai. [Q35, Q36, Q37, Q38]
- Isoneutronic / isotone nuclei mein neutron number same hota hai, proton number different. $^{14}C_6$, $^{15}N_7$ aur $^{16}O_8$ tino mein neutrons = 8, isliye ye isoneutronic group banate hain. [Q39]

11. Radioactivity — Detection & Discovery

- Geiger–Müller counter ionizing radiation ko detect/count karne ka instrument hai aur radioactivity measurement context mein commonly poocha jata hai. [Q40]
- Radioactivity ki discovery Henri Becquerel ne 1896 mein uranium salts ke saath experiments ke dauran ki thi. [Q41]

⚠ EXAM TRAP — Radioactivity — Instrument vs SI Unit — Q40

▸ Geiger–Müller counter radiation detect/count karne ka instrument hai. Radioactive activity ki SI unit Becquerel (Bq) hoti hai — instrument aur unit ko mix na karein. [Q40]